Membrane transport comparison

Compare pathway, driving force and energy coupling rather than protein use alone.

Mechanism	Pathway	Driving force	Energy link
Simple diffusion	Lipid bilayer	Down chemical gradient	No direct cellular energy input; e.g. O2.
Facilitated diffusion	Channel or carrier	Down electrochemical gradient	No direct energy input; pathway is selective.
Primary active transport	Pump protein	Coupled uphill transport	Direct energy coupling; the Na+/K+ pump uses ATP.
Secondary active transport	Cotransporter	Downhill solute drives uphill cargo	Uses stored gradient energy; not direct ATP hydrolysis by the cotransporter.
Vesicular transport	Membrane budding/fusion	Bulk uptake or release	Energy-dependent processes; endocytosis/exocytosis.

How an ion gradient can power uptake

ATP-powered pump builds Na+ gradient → Na+ enters downhill through cotransporter → Glucose moves uphill with Na+

For ions, combine concentration and voltage: the electrochemical gradient determines downhill movement. A transport protein alone does not make movement active. Blocking a pathway can prevent rapid transport even when a large gradient exists.

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Self-check questions and explanations. Refer to the visual guide on the front.

How do simple diffusion, facilitated diffusion and primary active transport differ?

Simple diffusion crosses lipids downhill. Facilitated diffusion uses a channel/carrier downhill. Primary active transport directly couples an energy source, such as ATP hydrolysis, to uphill transport.

Why can an ATP-coupled Na+ pump move ions uphill?

An ATP-powered Na+ pump can move Na+ uphill against its electrochemical gradient by coupling transport to ATP hydrolysis. Concentration and voltage both contribute to whether a particular ion movement is uphill.

Does using a transport protein always make transport active?

No. A carrier can move its substrate downhill without direct energy input. Active/passive classification depends on driving force and energy coupling, not whether a protein is present.

How can Na+ entry drive uphill glucose entry?

Na+ goes inward downhill along its electrochemical gradient; glucose goes inward uphill. The immediate energy source is the Na+ gradient, coupled through the cotransporter: secondary active transport.

Why can secondary transport continue briefly after the ATP-powered pump stops?

An existing Na+ gradient stores free energy and can drive brief continued cotransport. Without pumping, leaks and cotransport dissipate it. The cotransporter need not hydrolyze ATP directly to depend ultimately on ATP-driven pumping.

Does a glucose gradient ensure rapid crossing if all carriers are blocked?

Glucose crosses the bare lipid core very slowly. A concentration difference does not guarantee rapid transport: permeability and available pathways constrain flux. Do not claim that a blocked carrier reverses the gradient.